Algebra 1 Name_Answer Key________
Unit 6 Date____________________
Lesson 2 Homework Period__________________

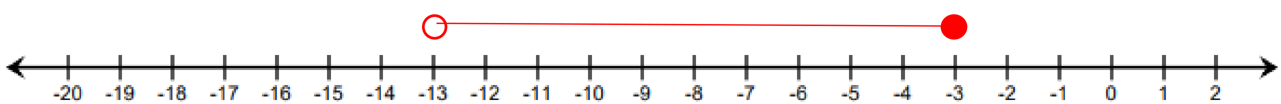
Use the following conjunctive inequality to answer questions 1-3.

$$5v + 10 \le -4v - 17 < 9 - 2v$$

Rewrite the compound inequality as two single inequalities and solve each one.

1. $5v + 10 \le -4v - 17$ $+4v \quad + 4v$ $9v + 10 \le -17$ $-10 \quad - 10$ $\frac{9v}{9} \le \frac{-27}{9}$ $v \le -3$	2. $-4v - 17 < 9 - 2v$ $+2v \quad + 2v$ $-2v - 17 < 9$ $+17 \quad + 17$ $\frac{-2v}{2} < \frac{26}{-2}$ $v > -13$

3. Graph the solution set of the compound inequality.



Kali just started a new sales floor job to save for college. She earns $15.75 per hour plus a flat fee of $50 per week. She wants to earn between $200 and $400 per week. The following inequality represents her earning potential:

$$200 \le 15.75x + 50 \le 400$$

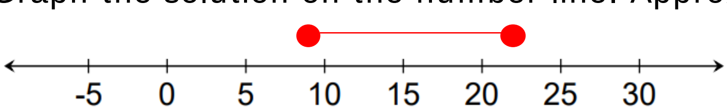
4. Solve the inequality.

$15.75x + 50 \ge 200$
$-50 \quad - 50$
$\frac{15.75x}{15.75} \ge \frac{150}{15.75}$
$x \ge 9\frac{11}{21}$

$15.75x + 50 \le 400$
$-50 \quad - 50$
$\frac{15.75x}{15.75} \le \frac{350}{15.75}$
$x \le 22\frac{2}{9}$

$9\frac{11}{21} \le x \le 22\frac{2}{9}$

5. Graph the solution on the number line. Approximate values as necessary.



6. Complete the statement: Kali can work between $9\frac{11}{21}$ and $22\frac{2}{9}$ hours each week.
≈ 9.5 ≈ 22.2

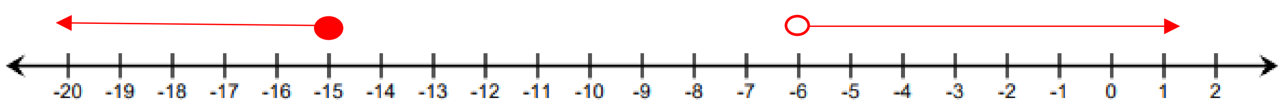
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{15}{4}\left(2x-\frac{2}{3}\right) < 64+\frac{11}{4}x \quad or \quad \frac{35x-11}{4} \leq -134$$

Solve each component of the compound inequality.

7. $-\frac{15}{4}\left(2x-\frac{2}{3}\right) < 64+\frac{11}{4}x$	8. $\frac{35x-11}{4} \leq -134$
$-7.5x + 2.5 < 64 + 2.75x$ $-2.75x \qquad\qquad - 2.75x$	$4 \cdot \frac{35x-11}{4} \leq -134 \cdot 4$
$-10.25x + 2.5 < 64$ $-2.5 \quad -2.5$	$35x - 11 \leq -536$ $+11 \quad +11$
$\frac{-10.25x}{-10.25} < \frac{61.5}{-10.25}$	$\frac{35x}{35} \leq \frac{-525}{35}$
$x > -6$	$x \leq -15$

9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

$$-20 \qquad -10 \qquad 0$$

-10 is not included because neither inequality includes the solution set.